Lab Report: Training dynamics of deep models

1 Introduction

In this report we compare several models on standard benchmarks. All experiments were run on a single GPU with 16 GB of memory. The dataset was split into 70 percent training data and the rest for testing.

2 Results

Figure 1 shows the accuracy during training.

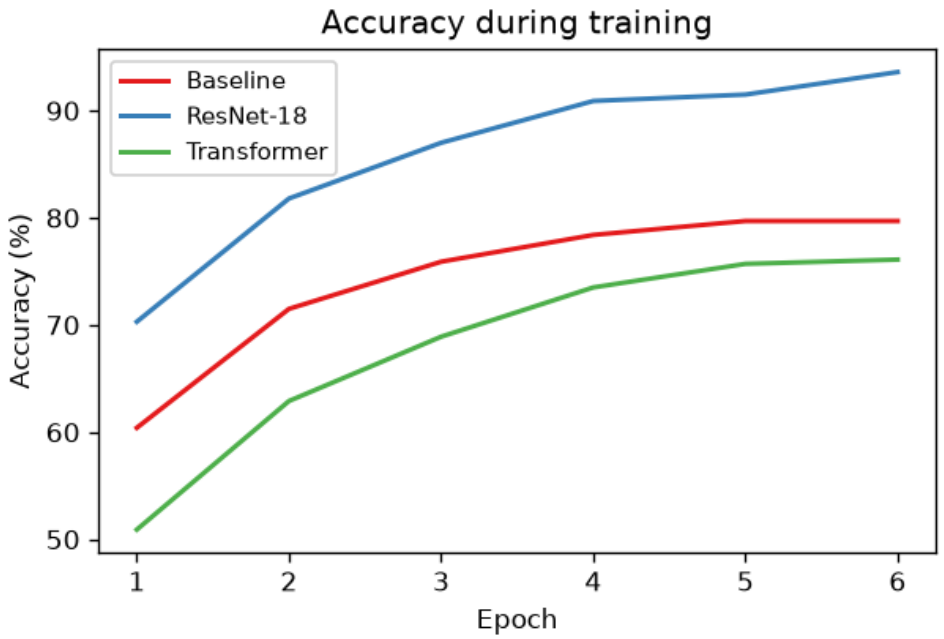


Figure 1: Accuracy during training for the models in this experiment.

Baseline outperforms Transformer by 2.3 percentage points at epoch 3. At epoch 3, ResNet-18 is 18 points better than Transformer.

Figure 2 shows the f1 score of MobileNet by dataset.

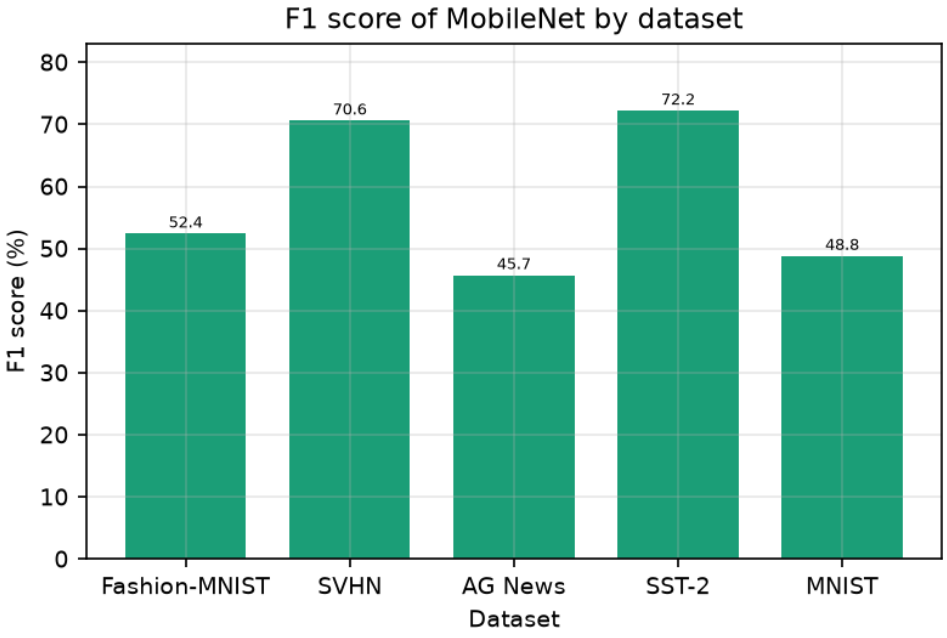


Figure 2: F1 score of MobileNet by dataset for the models in this experiment.

MobileNet reaches 70.6% F1 score on AG News. On MNIST, MobileNet obtains a F1 score of 72.2%. As shown in Figure 2, MobileNet reaches 70.5% F1 score on SST-2.

Figure 3 shows the validation accuracy of ViT-S by dataset.

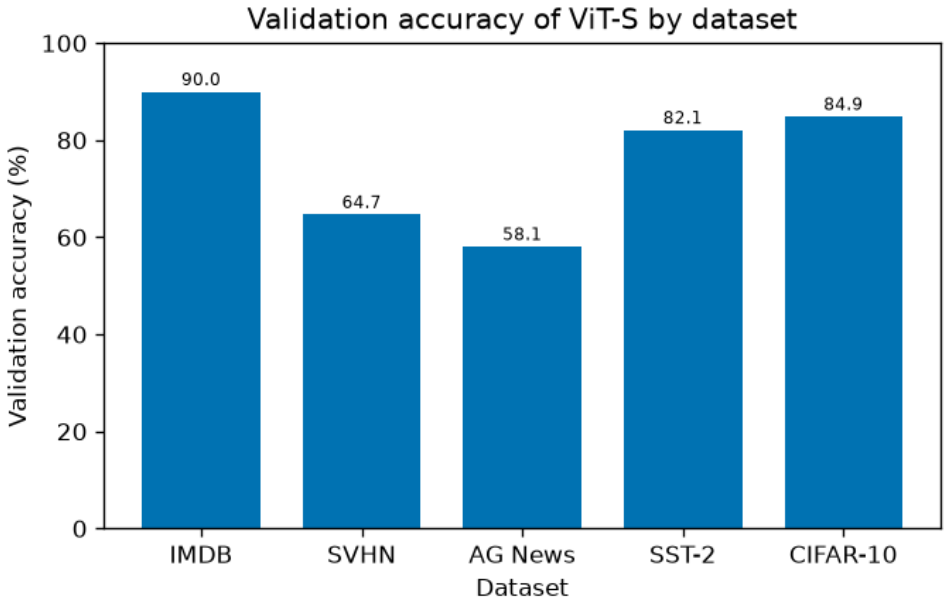


Figure 3: Validation accuracy of ViT-S by dataset for the models in this experiment.

ViT-S reaches 82.1% validation accuracy on SST-2. ViT-S scores 32 points higher on IMDB than on AG News.

Figure 4 shows the accuracy by dataset.

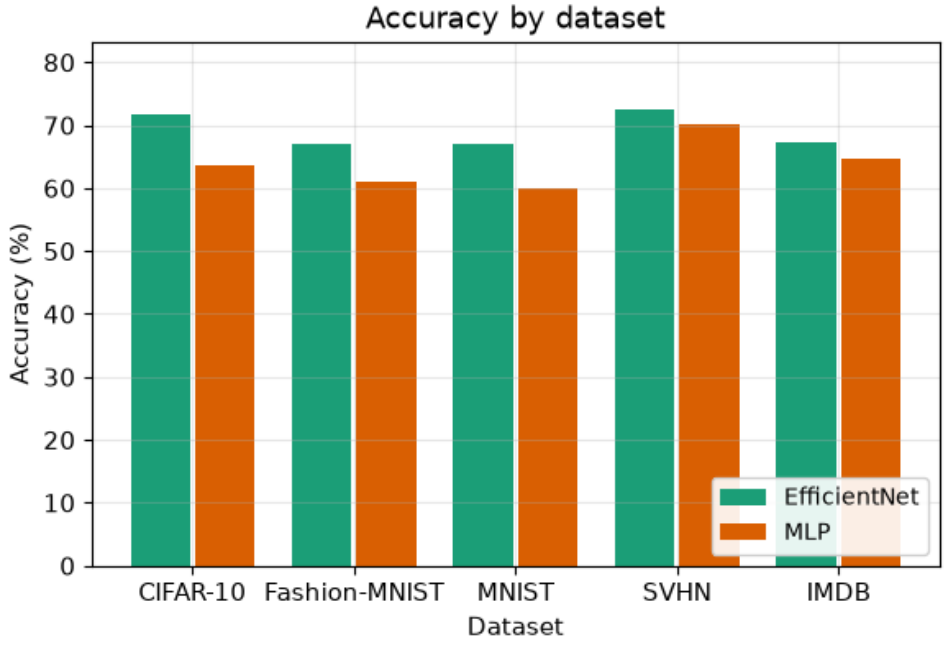


Figure 4: Accuracy by dataset for the models in this experiment.

As shown in Figure 4, EfficientNet achieves 61.0% accuracy on Fashion-MNIST. EfficientNet reaches 67.0% accuracy on Fashion-MNIST. On CIFAR-10, EfficientNet obtains a accuracy of 72%. MLP reaches 67.0% accuracy on Fashion-MNIST (Figure 4).

3 Discussion

Overall, the results are in line with what we expected from the literature.